Prediction Report: CV Delivery Forecasting for Workforce Planning

Case Study 2: Predictive Workforce Planning (CV Delivery Forecasting) Prepared for: RSGT Bangladesh – Terminal Operations **Report Date:** 10 April 2026
Analyst: Punam Das– Built on 5 months of historical CV delivery data (Aug 2025 – Jan 2026)

Executive Summary

Accurate forecasting of Covered Van (CV) deliveries is critical to balance labor and forklift allocation, minimize yard congestion, and control operational costs.

Using only the provided historical dataset (CV OCD Data Sep to Jan.csv), this report delivers a **simple, robust, and seasonality-aware forecast** for the first 10 days of February 2026.

Total forecasted CV deliveries (1–10 Feb 2026): 635 Average daily forecast: 63.5 (very close to the historical daily average of 65.3)

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1. Forecasted Daily CV Deliveries (1–10 February 2026)

Date	Day of Week	Forecasted CV Deliveries
2026-02-01	Sunday	**30**
2026-02-02	Monday	**73**
2026-02-03	Tuesday	**76**
2026-02-04	Wednesday	**66**
2026-02-05	Thursday	**84**
2026-02-06	Friday	**68**
2026-02-07	Saturday	**59**
2026-02-08	Sunday	**30**
2026-02-09	Monday	**73**
2026-02-10	Tuesday	**76**

Key Observation:

- Peak activity expected on **Thursday (84)** and **Tuesday (76)** – consistent with historical patterns.
- Lowest activity on **Sundays (30)** – typical weekend dip.

2. Methodology & Technical Note

Chosen Model: Day-of-Week Average (Seasonal Naive Forecast)

Step-by-step process:

1. Loaded the full historical CSV and filtered only **CV** records (CV /OCD == "CV").
2. Parsed delivery dates and aggregated daily CV delivery counts.
3. Created a complete daily time series (173 days from 12 Aug 2025 to 31 Jan 2026), filling missing days with 0 deliveries.
4. Restricted analysis to data **before 1 Feb 2026** (no data leakage).
5. Calculated the **average CV deliveries for each day of the week** across the entire 5-month period.
6. Mapped these historical averages to the corresponding weekdays in the forecast period (1–10 Feb 2026) and rounded to the nearest integer.

Historical Insights (5 months of actual data):

- Total historical days analyzed: **173**
- Overall average daily CV deliveries: **65.3**
- Day-of-week averages:
    - Thursday: 84.2
    - Tuesday: 76.0
    - Monday: 72.8
    - Friday: 68.0
    - Wednesday: 65.9
    - Saturday: 58.6
    - Sunday: 30.

3. Justification for the Chosen Method

- **Directly addresses the case study's key factors**: The model explicitly captures **Weekly Seasonality** (the strongest signal highlighted in the problem statement).
- **4-Day Rule & Yard Density Factor** are implicitly included because these operational realities are already reflected in the historical delivery volumes.
- **Simplicity and practicality**: No need for complex parameters or external software. The Terminal Manager can understand and use the forecast immediately for daily labor planning.
- **Why not other models?**
  - Simple Moving Average ignores weekly patterns.
  - Linear Regression or full Time-Series ML (SARIMA, Prophet, etc.) would overfit on only 5 months of data and add unnecessary complexity for a short 10-day horizon.
  - The day-of-week average is the most robust and interpretable choice given the data and business need.

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4. Assumptions (as permitted in the case study)

- The weekly delivery pattern observed from September 2025 to January 2026 will continue into early February 2026.
- No major external disruptions (public holidays, vessel delays, weather events, or port regulations changes) during 1–10 Feb 2026.
- The February import scenario numbers (157, 59, 0, 0, 0, 184, 319, 221, 87, 115) were noted but **not directly used**. Early-February deliveries are driven primarily by existing yard inventory under the 4-Day Rule; the historical pattern already accounts for typical import-to-delivery lag.
- The yard had 300 CV containers on 1 September 2025 (acknowledged but not required for this seasonality-based model).